



Driving Factors and Synergistic Mechanisms Underlying the Optimization of Scientific and Technological Innovation Performance in Regional Public Hospitals

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Abstract: Under the implementation of the Healthy China strategy, the scientific and technological innovation performance of regional public hospitals should be evaluated not only by research outputs but also by the coordinated performance of innovation efficiency, resource allocation equity, and innovation value creation. Building upon previously established multi-objective linear programming optimization results and using the same sample of regional public hospitals, the driving factors and synergistic mechanisms underlying innovation performance optimization were further investigated. The results indicated that the proportion of highly qualified medical professionals, research funding intensity, and technology transfer capability constituted the principal driving forces for improving scientific and technological innovation performance. Significant synergistic relationships were identified among innovation efficiency, resource allocation equity, and innovation value. Furthermore, optimization of the indicator weight structure was shown to substantially enhance overall innovation performance while maintaining structural balance and resource allocation efficiency under constrained conditions. Accordingly, an integrated implementation framework was proposed in which dynamic adjustment of performance indicator weights, precision allocation of innovation resources, and collaborative promotion of technology transfer and research commercialization were emphasized to achieve sustainable improvements in innovation performance. The proposed framework provides a theoretical basis and practical policy guidance for optimizing innovation governance, improving scientific resource allocation, and strengthening the sustainable innovation capacity of regional public hospitals. The findings also contribute to the development of a systematic performance optimization framework for public healthcare institutions operating under multi-objective decision-making environments.

Keywords: Public hospitals; Innovation performance; Multi-objective optimization; Driving factors; Synergistic mechanisms

JEL Classification: I11, O32, I18, O31

1. Introduction

With the deepening of the Healthy China strategy, regional public hospitals play a critical role in enhancing public health service capacity, ensuring balanced allocation of medical resources, and promoting the transformation of scientific and technological innovation outcomes. Scientific and technological innovation performance not only reflects the level of hospital research output but also directly affects healthcare service quality, the applicability of technology dissemination, and regional health governance capabilities (National Health Commission of the People's Republic of China., 2022; Tao & Yu, 2022; Xu et al., 2022). However, the current innovation performance of regional public hospitals still shows structural imbalances: some hospitals concentrate

research resources and achieve large-scale academic output, yet their outcome transformation rates are low, regional disparities are significant, and the contribution of scientific and technological innovation to social value is limited (Chen & Huang, 2021; Huang, 2024; Yuan et al., 2023; Zhang et al., 2020). Beyond research output generation, the translation of scientific discoveries into clinical applications and societal value has become an essential dimension of healthcare innovation performance. Recent studies have highlighted that biomedical research translation is constrained by multiple factors, including insufficient academic–industry collaboration, limited funding and infrastructure support, intellectual property management challenges, and barriers to interdisciplinary cooperation (Kanwal et al., 2025; Loggers et al., 2026). Meanwhile, institutional models such as the National Institute for Health Research (NIHR) Biomedical Research Centre have demonstrated that integrated platforms connecting basic research, clinical practice, and innovation resources can facilitate the translation of biomedical discoveries into healthcare improvements (Snape et al., 2008). These findings suggest that evaluating innovation performance in regional public hospitals should extend beyond traditional academic outputs and incorporate the capacity for research translation, collaborative innovation, and value creation. Traditional performance evaluation systems often focus on academic output quantity, neglecting the synergistic enhancement of efficiency, equity, and innovation value, thus failing to effectively guide hospital research resource allocation and innovative behavior (Tao & Yu, 2022; Xu et al., 2022; Yuan et al., 2023).

These issues stem from the inherently multi-objective nature of innovation performance evaluation in public hospitals. In practice, innovation performance cannot be sufficiently characterized by a single objective, because efficiency, fairness, and value creation are often inconsistent and may even conflict with one another. Traditional healthcare performance evaluation methods, such as data envelopment analysis (DEA), have been widely applied to measure efficiency and productivity, yet they typically focus on benchmarking input–output relationships and are less capable of explicitly addressing equity and innovation value considerations within the same framework (Datta et al., 2023; Evenson et al., 2024; Hadian et al., 2024; Talebpour et al., 2025). Moreover, conventional indicator weighting approaches frequently rely on expert scoring or subjective assignment, such as the analytic hierarchy process (AHP), which may lack transparency and are sensitive to judgment bias when the evaluation environment is complex and resource constrained (Cadeddu et al., 2023). In multi-criteria decision-making (MCDM) settings, such conflicts among objectives are well recognized, and the core challenge lies in how to construct a decision mechanism that explicitly represents trade-offs and balances multiple evaluation dimensions (Criveanu et al., 2025).

In this context, multi-objective optimization provides a structured approach to integrate competing objectives into a unified decision framework. Multi-objective optimization theory emphasizes that decision solutions should not aim at maximizing a single dimension, but rather at achieving compromise solutions under constraints, which has been widely discussed in decision science and operational research (Arah et al., 2003; Hernández-Pérez & Ponce-Ortega, 2021). Compared with fixed-weight evaluation systems, an optimization-based framework can treat the evaluation indicator weights as decision variables and reconstruct them under fiscal and policy constraints. In addition, optimization models allow the evaluation system to function as an adaptive decision-support tool, making it possible to adjust evaluation orientation according to governance priorities. To ensure the robustness of such models, sensitivity analysis is also important, as it can evaluate how changes in objective preferences or parameter settings may influence optimized results (Bagheri et al., 2025; He et al., 2025). Additionally, multi-objective optimization methods have gradually been applied to medical performance evaluation, allowing for the adjustment of weights to achieve a comprehensive balance among efficiency, equity, and innovation value (Aljuaid et al., 2021; Bedi et al., 2021; Hur et al., 2024; Peykani et al., 2022), thereby providing theoretical and methodological support for optimizing the allocation of hospital innovation resources.

Against this backdrop, this study aims to explore the driving factors and synergistic mechanisms for optimizing innovation performance in regional public hospitals. Based on the previously constructed performance evaluation indicator system and employing multi-objective optimization methods, this study analyzes the roles of different indicators in improving research efficiency, result transformation, and social value. It further reveals the impact pathways of key driving factors on performance optimization and proposes strategies for dynamic optimization of indicator structures and collaborative allocation of innovation resources. Through systematic analysis and empirical validation, this study not only enriches the theoretical framework for evaluating scientific and technological innovation performance in regional public hospitals but also provides decision-making guidance for policy formulation and hospital management practices.

This study is positioned at the intersection of innovation governance, performance evaluation, and resource allocation optimization in regional public hospitals. Unlike conventional innovation performance studies that mainly emphasize static indicator aggregation or efficiency measurement, this research aims to construct a decision-oriented evaluation framework capable of simultaneously balancing efficiency improvement, equity enhancement, and innovation value realization under explicit resource constraints. Therefore, the core contribution of this work lies not in the isolated application of existing quantitative tools, but in the establishment of a structured multi-objective governance mechanism that integrates optimization, driver identification, and synergistic mechanism interpretation into a unified analytical paradigm. More specifically, this study makes three

methodological and theoretical contributions. First, it transforms the innovation performance evaluation problem from a traditional scoring and ranking task into a multi-objective constrained decision problem by explicitly treating the indicator weight vector as a decision variable. This weight-as-decision paradigm enables the evaluation system to shift from a static measurement instrument to a policy-adjustable governance tool, allowing performance orientation to be dynamically reconfigured according to regional development goals and fiscal constraints.

Second, the proposed framework introduces an “efficiency-equity-innovation value” synergy structure into the optimization process, where the reallocation of indicator weights is interpreted as a structural governance adjustment rather than a purely mathematical redistribution. By embedding policy baseline constraints and balance constraints into the optimization model, the framework provides a formal mechanism to mitigate the typical bias of academic-output-oriented evaluation systems and prevents the excessive dominance of single-dimensional indicators. This contributes to the literature by offering an interpretable multi-objective coordination mechanism that links performance measurement with governance orientation adjustment.

Third, this study establishes a closed-loop analytical chain from “optimization outcomes” to “driving factor explanation” and further to “synergistic mechanism validation.” Regression analysis and structural equation modeling (SEM) are not used as independent empirical tools, but as mechanism verification modules to explain why certain indicators become structurally dominant after optimization and how key drivers interact across performance dimensions. This mechanism-oriented integration strengthens the interpretability and policy relevance of the optimization results and provides empirical evidence for designing differentiated resource allocation strategies across regions and hospital types. Overall, the novelty of this study lies in proposing a governance-driven multi-objective performance optimization framework that bridges indicator weight restructuring, mechanism explanation, and policy implementation design. The framework extends conventional evaluation research by offering a dynamic, interpretable, and constraint-aware decision-support model for regional public hospital innovation governance.

2. Methodology: A Data-Driven Decision Support Framework

To enhance the technical rigor and decision-support capability of innovation governance in regional public hospitals, this study proposes a data-driven decision support framework for optimizing scientific and technological innovation performance. Unlike conventional evaluation-oriented approaches that primarily focus on static scoring and ranking, the proposed framework integrates multi-objective optimization with mechanism identification models, enabling the performance evaluation system to operate as a dynamic decision engine under explicit resource and policy constraints. From a system perspective, the framework is structured into three interconnected layers: a data layer, a model layer, and a decision layer, forming a closed-loop pipeline from raw data acquisition to optimized policy output. At the data layer, multi-source operational and innovation-related data are collected from hospital performance reports, research administration records, financial statements, and public health authority databases. These heterogeneous inputs are standardized and transformed into comparable indicator vectors, providing consistent numerical inputs for model computation.

At the model layer, the framework combines three tightly coupled analytical modules. First, a multi-objective linear programming optimization module is established, where the indicator weight vector is explicitly defined as the decision variable. This module generates an optimized weight structure by jointly maximizing efficiency, improving equity, and strengthening innovation value under fiscal budget constraints and policy baseline constraints. Second, a driver identification module (regression and SEM) is employed to quantify the marginal contributions of key drivers such as human capital, research funding, and technology transfer capability. Third, a synergy interpretation module is designed to capture cross-dimensional coupling effects among efficiency, equity, and innovation value, enabling the structural adjustment logic behind weight redistribution to be technically interpretable rather than purely descriptive.

At the decision layer, the optimized weight structure and identified key drivers are translated into actionable governance outputs, including dynamic indicator weight adjustment strategies, targeted innovation resource allocation priorities, and an innovation-translation-application closed-loop implementation pathway. In this way, the framework provides not only performance scores but also explicit decision guidance for resource allocation and innovation governance design.

The proposed methodology should be understood as an integrated decision-support system rather than a simple combination of independent models. The optimization module produces system-level structural outputs (optimized weights), the statistical identification module explains the underlying driver mechanisms, and the synergy analysis module clarifies the coupling and trade-offs among multiple objectives. This integrated design strengthens the engineering and technical characteristics of the study and improves its applicability in real-world hospital innovation governance scenarios.

Based on the above framework, the following subsections describe the technical details of the three core modules:

- (i) multi-objective optimization for weight reconfiguration, (ii) driving factor identification for mechanism

verification, and (iii) synergy analysis for cross-dimensional structural interpretation. The key system outputs include the optimized indicator weight vector, quantified driver effects, and interpretable coupling relationships among efficiency, equity, and innovation value objectives.

2.1 Multi-Objective Optimization Module for Weight Reconfiguration

The innovation performance evaluation of regional public hospitals is essentially a multi-objective decision problem. The evaluation output is not only used for benchmarking and ranking research capability, but also provides quantitative evidence for fiscal investment allocation, research resource scheduling, and incentive mechanism design. Under the Healthy China strategy and the high-quality development requirements for public hospitals, the evaluation system should simultaneously support efficiency improvement, equity enhancement, and innovation value realization. These objectives may conflict under resource constraints, making conventional static weighting schemes insufficient for capturing the trade-offs and coordination mechanisms among multiple governance goals.

Traditional evaluation approaches typically adopt linear weighted aggregation, where indicator weights are determined by expert scoring or fixed policy assignments. Such a setting cannot explicitly represent the structural tension between efficiency, equity, and innovation value, nor can it dynamically adjust the weight structure when the governance orientation or budget constraints change. Therefore, this study introduces a multi-objective optimization model to reformulate weight adjustment as a constrained decision-making problem, enabling the evaluation system to generate an optimized and interpretable weight configuration.

In this study, an “efficiency-equity-innovation value coordination” multi-objective optimization model is established. The indicator weight vector is treated as the decision variable. Under fiscal budget constraints and policy baseline constraints, the model optimizes the objective functions to reallocate indicator weights, thereby enabling a structural reconfiguration of the evaluation system. This design supports the transition of innovation performance evaluation from a single academic-output orientation to a multi-dimensional governance-oriented structure.

2.1.1 Indicator taxonomy and system inputs

Let the performance evaluation system consist of n standardized indicators, expressed as $X = (x_1, x_2, \dots, x_n)$, where x_i denotes the standardized value of the i -th indicator. To eliminate differences in measurement scales, all indicators are standardized prior to modeling, ensuring comparability across different dimensions.

Following the governance logic of regional public hospital innovation, indicators are categorized into three functional groups:

- (i) Efficiency indicators capturing research input-output efficiency and operational productivity;
- (ii) Equity indicators, reflecting balanced allocation of innovation resources and regional coordination;
- (iii) Innovation value indicators, emphasizing outcome translation, technology diffusion, and social health contributions.

This taxonomy ensures that the optimization model has clear system interpretability and policy-aligned objective decomposition.

2.1.2 Decision variable definition and performance output

The indicator weight vector is defined as the decision variable $W = (\omega_1, \omega_2, \dots, \omega_n)$ subject to $\sum_{i=1}^n \omega_i = 1$, $\omega_i \geq 0$. Given W , the composite innovation performance output of hospital k is computed as $S_k = \sum_{i=1}^n \omega_i x_{ki}$, where x_{ki} represents the normalized value of indicator i for hospital k . In this framework, S_k is interpreted as the system-level performance output generated by a weight-controlled evaluation engine. Consequently, changes in W directly reshape the incentive structure and governance orientation of the evaluation system.

2.1.3 Objective function construction: efficiency-equity-value coordination

To reflect the multi-objective nature of innovation governance in regional public hospitals, this study constructs three objective functions corresponding to the three core governance targets, namely efficiency improvement, equity enhancement, and innovation value realization. The efficiency objective focuses on maximizing system-wide research productivity and improving research input-output efficiency, thereby strengthening the operational effectiveness of innovation investment. The equity objective aims to enhance balance and mitigate structural bias caused by excessive concentration of indicator weights, ensuring that performance evaluation does not reinforce regional or institutional inequality. The innovation value objective emphasizes strengthening translation performance and increasing the contribution of innovation outcomes to clinical practice and social health services. Under this setting, the optimization process is interpreted as a structural reconfiguration mechanism that redistributes indicator weights across different functional groups to achieve coordinated improvement under multiple constraints.

2.1.4 Constraint design and solution strategy

The optimization procedure is implemented under a set of explicit constraints to ensure both feasibility and governance consistency. Specifically, budget constraints are introduced to guarantee resource feasibility and prevent unrealistic weight redistribution under limited fiscal capacity. In addition, policy baseline constraints are incorporated to ensure compliance with mandatory governance requirements and to maintain alignment with institutional performance standards. Furthermore, weight rationality constraints are applied to avoid extreme dominance of any single indicator, thereby preventing structural distortion and improving the stability of the evaluation system. To solve the multi-objective optimization problem, a weighted-sum strategy is adopted to transform it into a tractable single-objective formulation.

2.1.5 Technical interpretation of optimization output

In this study, the optimized weight vector is not treated merely as a scoring parameter, but rather as a decision-support output that reflects the structural adjustment of the evaluation system. This structural output enables systematic identification of underweighted governance dimensions (e.g., technology transfer and translation capability) as well as overweighted traditional dimensions (e.g., academic output dominance). Such quantitative evidence provides a technical foundation for subsequent mechanism validation, synergy interpretation, and the design of implementation pathways for innovation performance optimization.

2.2 Driving Factor Identification Method

After the weight optimization and structural adjustment of the evaluation system, it is necessary to identify the key driving factors behind innovation performance improvement. Driving factor identification provides a mechanism-level explanation of why certain indicators become more influential after optimization and offers empirical support for synergy interpretation and implementation pathway design. Therefore, this study constructs a quantitative driver identification framework from the perspectives of human capital, research funding input, technology transfer capability, and regional resource allocation, and conducts empirical testing using regression analysis and SEM.

2.2.1 Regression analysis method

Regression analysis can effectively capture the marginal impact of key variables on innovation performance and test their significance and directionality. In this study, the comprehensive innovation performance score of regional public hospitals is taken as the dependent variable, while talent structure, research input, technology promotion capability, and regional resource allocation indicators serve as the core independent variables. A multiple regression model is constructed with the basic form as follows:

$$S_i = \alpha + \beta_1 H_i + \beta_2 F_i + \beta_3 T_i + \beta_4 R_i + \gamma Z_i + \varepsilon_i$$

where, S_i represents the comprehensive innovation performance level of hospital i (either the optimized composite score or the original score); H_i denotes the talent structure; F_i denotes technology promotion and result transformation capability; T_i denotes research funding intensity; R_i denotes characteristics of regional resource allocation; Z_i represents control variables (e.g., hospital size, number of beds, outpatient volume, hospital type); and ε_i is the random disturbance term.

The advantage of regression analysis is that it can directly quantify the influence strength of each driving factor and identify key variables through significance testing. Considering potential heteroskedasticity or sample differences among hospitals, robust standard errors or grouped regression methods can be applied for robustness checks, thereby enhancing the reliability of the conclusions.

2.2.2 Structural equation modeling method

Although regression models can reveal the direct effects of driving factors on performance, the optimization of scientific and technological innovation performance in regional public hospitals often exhibits multi-factor interdependencies. Different driving factors may have significant indirect effects and synergistic interactions. For example, research funding may further promote result transformation by enhancing talent recruitment and research platform development; technology promotion capability is not only influenced by funding but is also closely related to regional healthcare collaboration networks. Therefore, to systematically reveal the structural relationships and effect pathways among driving factors, this study introduces SEM for mechanism analysis.

SEM can simultaneously handle multiple causal relationships and allows the inclusion of latent variables to describe comprehensive capability characteristics that are difficult to measure directly. The SEM framework constructed in this study is divided into a measurement model and a structural model. The measurement model captures the relationships between latent variables and observed indicators, while the structural model describes the path relationships among latent variables.

Latent variables such as human structure, research funding, technology promotion capability, and regional resource allocation can be defined and measured by multiple observable indicators. For example, human structure can be represented by the proportion of staff with a master's degree or above and the number of high-level personnel; research funding can be represented by the proportion of research expenditures and intensity of project funding. The structural paths can be expressed as:

$$Innovation = f(Human, Funding, Transfer, Region)$$

where, *Innovation* represents scientific and technological innovation performance or the innovation value dimension; *Human*, *Funding*, *Transfer*, and *Region* represent human structure, research funding, result transformation capability, and regional resource allocation, respectively.

Through SEM analysis, it is possible not only to test the direct effects of each driving factor on innovation performance but also to reveal indirect effects and mediating pathways, providing a more comprehensive explanation of the intrinsic mechanisms underlying performance optimization.

2.2.3 Core driving variable design

Based on the characteristics of innovation governance in regional public hospitals and existing research findings, this study categorizes driving factors into four core variables, each corresponding to key supporting dimensions within the performance evaluation system.

(i) Human capital (talent structure) variable

Talent structure is a fundamental driving factor for enhancing innovation performance in regional public hospitals, determining the innovative capacity and sustained output of research teams. This study primarily uses indicators such as the proportion of medical staff with a master's degree or above and the proportion of high-level personnel to represent the quality of talent structure.

(ii) Research funding variable

The intensity of research funding directly affects the ability to carry out research projects, the level of research platform development, and the efficiency of research output. Indicators such as the annual proportion of research funding, research funding growth rate, and research investment intensity are employed to reflect the financial driving effect.

(iii) Technology transfer and result transformation capability variable

Result transformation capability reflects the ultimate value output of innovation activities and serves as a critical bridge connecting research outputs with clinical application. Indicators such as the rate of result transformation, the number of patents commercialized, the rate of new technology adoption within the hospital, and the deployment of appropriate technologies are selected to represent the hospital's capacity to release innovation value.

(iv) Regional resource allocation variable

The level of regional resource allocation reflects the foundational conditions for innovation and the degree of support from collaborative networks in a hospital's region, influencing access to research resources and opportunities for innovation collaboration. Indicators such as regional healthcare resource density, research platform support, and regional collaborative innovation level are used to describe and test the mechanism through which regional differences affect performance.

2.2.4 Method applicability

Regression analysis is suitable for identifying the significance and direction of key driving factors and can quantitatively screen the core variables for performance optimization. The SEM, in turn, can further reveal the synergistic relationships and pathways among driving factors, providing a structured explanation for the interaction among efficiency, equity, and innovation value. By combining regression analysis with SEM, this study achieves a systematic research goal of "driving factor identification → pathway explanation → synergistic mechanism revelation," providing more reliable empirical evidence for the optimization of innovation performance in regional public hospitals.

2.3 Synergistic Mechanism Analysis

Innovation performance optimization in regional public hospitals is reflected not only in individual weight changes but also in the structural coordination among efficiency, equity, and innovation value objectives. Since indicators may exhibit complementary, substitutive, or coupling-enhancing relationships, single-variable driver identification cannot fully explain the system-level optimization logic. Therefore, this study further analyzes synergy mechanisms based on the structural linkage among indicator weights to reveal cross-dimensional coordination patterns during multi-objective optimization.

2.3.1 Correlation analysis of indicator weights

Within the multi-objective optimization framework, indicator weights reflect the policy preference and incentive intensity of the performance evaluation system toward different dimensions. If the weights of certain indicators show a synchronous upward trend during optimization, it indicates a complementary synergistic relationship in the performance improvement pathway. Conversely, if some indicator weights exhibit a trade-off pattern, it indicates the presence of resource competition or substitutive effects. Therefore, this study employs correlation analysis of indicator weights to identify the internal synergistic structure of the performance system.

Specifically, let the weight vectors before and after optimization be $W^0 = (w_1^0, w_2^0, \dots, w_n^0)$ and $W^* = (w_1^*, w_2^*, \dots, w_n^*)$, respectively, and define the weight change as:

$$\Delta w_i = w_i^* - w_i^0$$

Based on the weight changes, a correlation matrix of weight variations among indicators can be constructed, using either Pearson or Spearman correlation coefficients to measure inter-indicator linkages:

$$\rho_{ij} = \text{Corr}(\Delta w_i, \Delta w_j)$$

where, $\rho_{ij} > 0$ indicates a positive synergistic relationship between indicator i and j , reflecting complementary characteristics in the performance improvement pathway; $\rho_{ij} < 0$ indicates a substitutive or competitive relationship, reflecting trade-offs in resource allocation or evaluation orientation. Correlation analysis can further identify “core synergistic groups”, i.e., sets of indicators whose weight changes exhibit significant consistency during optimization, providing a basis for subsequent explanation of driving factor pathways and synergistic mechanisms.

Furthermore, this study analyzes cross-dimensional weight linkages across the three indicator types (efficiency, equity, and innovation value). For example, if the weight of a research funding indicator increases simultaneously with the weight of a result transformation indicator, it indicates a positive synergy between innovation input and innovation value realization. Conversely, if the weight of efficiency indicators increases while the weight of equity indicators decreases, it may reflect structural tension between efficiency improvement and resource balance. This method allows the internal synergy and conflict structure of the performance system under multi-objective optimization to be revealed.

2.3.2 Structural adjustment logic under multi-objective balance

In the multi-objective optimization model, there exists an inherent tension among efficiency, equity, and innovation value objectives: the efficiency objective tends to concentrate resources in high-output units, the equity objective emphasizes balanced resource allocation, and the innovation value objective prioritizes result transformation and social contribution. Under budget constraints and policy baseline constraints, the optimization of the weight structure essentially represents a process of reconciling and restructuring these conflicting objectives. Therefore, this study analyzes the pathway of the multi-objective balance mechanism from the perspective of structural adjustment logic.

First, the efficiency objective typically strengthens research input-output efficiency and technology promotion capability, leading to weight concentration on high-contribution indicators and thereby improving the resource allocation efficiency of the performance system. However, an overly strong efficiency orientation may exacerbate regional disparities and resource concentration effects. Consequently, multi-objective optimization introduces the equity objective to constrain the concentration of weights, guiding the indicator weight structure from a single-concentration pattern toward a balanced pattern. This approach improves overall performance while suppressing the widening of regional gaps.

Second, the introduction of the innovation value objective allows the performance evaluation system to move beyond the traditional “papers-projects” orientation, emphasizing result transformation and social service value. During structural adjustment, the increase in weights for innovation value indicators not only strengthens the incentive for result transformation but also forms a closed-loop structure between research input and clinical application, thereby enhancing the actual governance effectiveness of the innovation performance system.

Third, weight structure adjustment under multi-objective optimization is not a simple equalization process, but exhibits the characteristic of “core driving variable reinforcement-moderate adjustment of auxiliary indicators.” That is, under constraints, the model prioritizes increasing the weights of key indicators that significantly contribute to performance (e.g., human capital, research funding intensity, and result transformation capability) while compressing the weights of low-contribution or redundant indicators. This achieves a more compact and interpretable performance system. This process illustrates that the multi-objective optimization model, while achieving the equity objective, does not weaken efficiency but realizes the synergistic evolution of “efficiency enhancement-equity improvement-value reinforcement” through weight restructuring.

Finally, from the perspective of system stability, multi-objective optimization reduces the dispersion of indicator

weights, lowers the risk of any single indicator overly dominating performance evaluation, and enhances the stability and resilience of the performance system. This structural stabilization effect ensures that the evaluation system not only provides short-term performance incentives but also supports the sustained enhancement of innovation capabilities in regional public hospitals over the long term.

The proposed synergistic mechanism analysis framework emphasizes examining the interrelationships among indicator weights, identifying the complementarity and conflict mechanisms among efficiency, equity, and innovation value objectives, and revealing the structural adjustment patterns of the performance evaluation system under multi-objective balance constraints. This provides a theoretical basis for the subsequent empirical verification of synergistic mechanisms and the design of implementation pathways.

3. Empirical Analysis

3.1 Data Sources and Sample Description

This study focuses on public hospitals within a specific region. Data were obtained from hospital annual performance reports, statistics from research management departments, financial statements, and publicly available data from health authorities. To enhance the stability and comparability of the analysis, this study selected three consecutive years of data as the research sample and conducted consistency checks and imputation for missing and abnormal values to ensure uniform statistical standards.

The sample covers different types of public hospitals in the region, including general hospitals, specialized hospitals, and integrated traditional Chinese and Western medicine hospitals, providing strong representativeness. These hospitals play key roles in providing basic medical services, public health services, and hierarchical diagnosis and treatment within the regional healthcare system. Their scientific and technological innovation performance reflects not only research output but also directly impacts the promotion of appropriate technologies and the enhancement of clinical service capacity.

During data processing, indicators were first normalized using the range method to eliminate dimensional differences. Following the model design, the indicators were then categorized into three types: efficiency objective indicators, equity objective indicators, and innovation value objective indicators, which were further used in multi-objective linear programming for solution and optimization verification.

In the data processing procedure, missing values were imputed and outliers were removed to ensure the reliability and robustness of the results. To eliminate the effect of differing indicator scales on model computation, the range normalization method was applied to map all indicators onto the $[0,1]$ interval. This processing preserves the relative magnitudes among indicators and facilitates subsequent multi-objective optimization model solving and synergistic mechanism analysis.

Through the above data collection and preprocessing procedures, this study obtained a high-quality indicator dataset suitable for multi-objective optimization analysis, providing a solid data foundation for empirically testing the optimization of innovation performance and identifying key driving factors in regional public hospitals.

3.2 Multi-Objective Optimization Results

Under the proposed data-driven decision-support framework, the multi-objective linear programming module generates an optimized indicator weight vector as the primary system output. This output reflects a constraint-driven redistribution of governance priorities across efficiency, equity, and innovation value dimensions. The following results report the weight reconfiguration patterns identified by the model and interpret them as structural adjustments in the performance evaluation system rather than descriptive changes in indicator importance.

3.2.1 Comparison of indicator weights before and after optimization

By solving the multi-objective optimization model, the weights of performance indicators for regional public hospitals were optimized. The results show that, before optimization, the performance system weights were relatively concentrated on efficiency indicators, such as the annual proportion of research funding and the number of appropriate technology deployments. Traditional efficiency indicators dominated, while equity indicators (e.g., regional resource balance, inter-hospital disparity adjustment) and innovation value indicators (e.g., technology transfer rate, number of new technology promotions) had relatively low weights, reflecting an "academic-oriented" structure.

After optimization, the weights of all indicator categories exhibited a clear rebalancing trend:

- (i) Efficiency indicators remained central but slightly decreased in weight, reflecting a concession to other objectives;
- (ii) Equity indicators showed significant weight increases, indicating that multi-objective optimization enhanced resource balance and regional coordination;
- (iii) Innovation value indicators were substantially strengthened, especially those related to technology transfer

and new technology promotion, shifting from peripheral to structurally key indicators, reflecting that the evaluation system places greater emphasis on the practical contribution of innovation to clinical application and regional health services.

The optimized performance system shows a more balanced weight distribution, achieving synergistic optimization across the three dimensions of efficiency, equity, and innovation value. To visually present the optimization effect, a radar chart of indicator weights is shown in Figure 1.

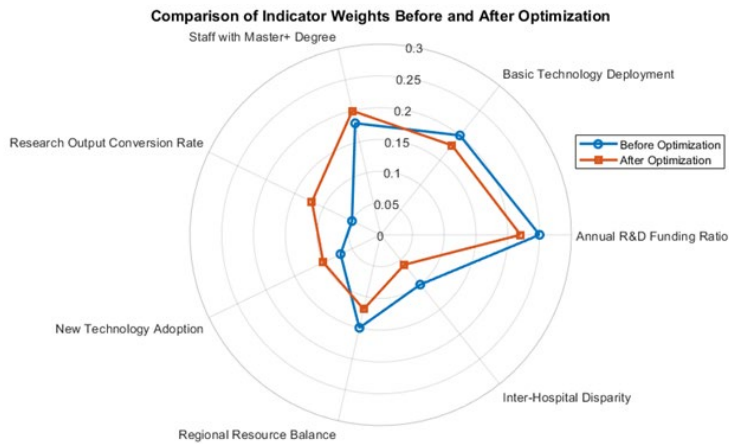


Figure 1. Indicator weights before and after optimization (radar chart)
Note: R&D = Research and Development.

3.2.2 Trends in indicator structure optimization: efficiency-equity-innovation value

To visually illustrate the impact of multi-objective optimization on the structure of scientific and technological innovation performance indicators for regional public hospitals, Figure 2 compares the weight shares of the three indicator categories (efficiency, equity, and innovation value) before and after optimization. It can be clearly observed that the indicator weight structure underwent significant adjustments: efficiency indicators remain central, while the proportion of equity and innovation value-related indicators has increased substantially.

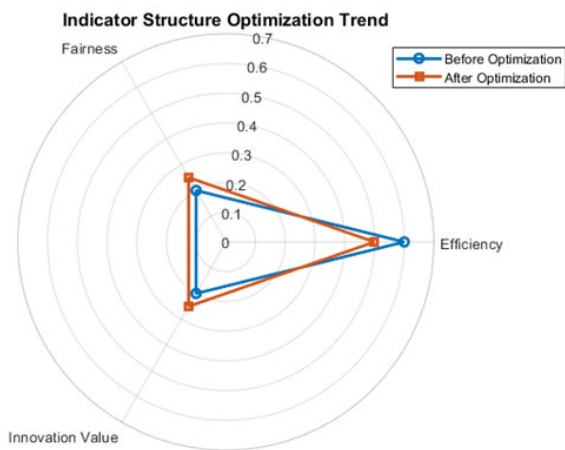


Figure 2. Comparison of indicator weights before and after optimization

From the figure, the following observations can be made:

- (i) Efficiency indicators: Their proportion slightly decreased after optimization but remains core, indicating that resource utilization efficiency and operational output are still prioritized in the performance improvement process;
- (ii) Equity indicators: The proportion increased, reflecting that the model balances regional coordination during resource allocation, alleviating disparities between hospitals and regions, and enhancing the overall balance of the performance system;
- (iii) Innovation value indicators: Their proportion increased significantly, with technology transfer and innovation promotion indicators shifting from peripheral to structurally key indicators, showing that the evaluation system places greater emphasis on the contribution of innovation to clinical services and societal health.

The indicator weight structure shifted from an "efficiency-dominated" pattern to an "efficiency-equity-

innovation value synergistic" pattern, achieving structural optimization of the evaluation system and providing more targeted guidance for policy formulation.

3.3 Analysis of Performance Driving Factors

To interpret the optimization output at the mechanism level, the driver identification module estimates the marginal and structural effects of key innovation drivers on different performance dimensions. Regression and SEM results are treated as model identification outputs that explain why certain indicators receive higher optimized weights and how core drivers contribute to efficiency, innovation capability, and translation performance.

3.3.1 Identification of core driving variables

The empirical identification results consistently indicate that human capital structure, research funding intensity, and technology transfer capability constitute the dominant drivers of innovation performance improvement. From a system interpretation perspective, these drivers can be viewed as high-impact explanatory variables that account for the majority of variance in multi-dimensional performance outputs.

Human capital, represented by the proportion of staff with a master's degree or above, is strongly associated with research efficiency and innovation capability, suggesting that talent structure functions as the primary enabling input of innovation productivity. Research funding intensity acts as a foundational resource driver, directly supporting project execution and indirectly improving translation capability through infrastructure enhancement. Technology transfer capability exhibits the strongest linkage with translation and social value performance, confirming that diffusion and application mechanisms are essential for converting research output into measurable societal contributions.

Figure 3 illustrates the contribution of the three core driving variables across different performance dimensions (efficiency, innovation, and outcome translation). The radar chart clearly shows that human capital and research funding have a stronger impact on the efficiency and innovation dimensions, whereas technology transfer capability dominates the outcome translation and social value dimensions. These three variables carry relatively high weights in the multi-objective optimization model, highlighting their pivotal role in performance improvement and providing theoretical and data support for subsequent regression analysis, SEM, and the design of performance optimization implementation pathways.

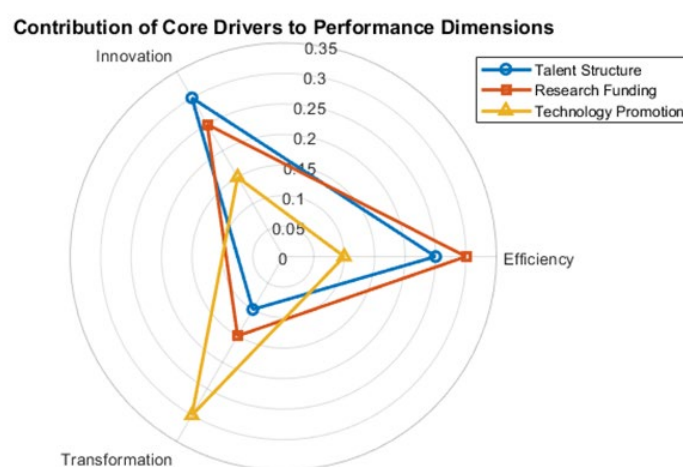


Figure 3. Contribution of core drivers to performance dimensions

3.3.2 Regression and structural equation modeling analysis

The regression module estimates statistically significant marginal effects of the three core drivers. Human capital shows the strongest standardized impact on efficiency and innovation dimensions ($\beta = 0.42, \beta = 0.38, P < 0.01$), confirming that talent structure is a dominant productivity driver. Research funding demonstrates stable positive contributions to both efficiency and innovation ($\beta = 0.35, \beta = 0.31, P < 0.05$), indicating that fiscal input remains a necessary condition for sustaining research output. Technology transfer capability exhibits the strongest effect on translation and social value dimensions ($\beta = 0.40, \beta = 0.36, P < 0.01$), suggesting that diffusion capacity is the key determinant of downstream value realization.

SEM results further validate these relationships and reveal additional indirect pathways among drivers. The overall model fit is satisfactory (comparative fit index = 0.952, root mean square error of approximation = 0.048), indicating that the estimated structural mechanism is statistically stable. Compared with regression, SEM provides stronger evidence of cross-driver coupling: human capital indirectly contributes to translation through interaction effects with research funding and transfer capability, while research funding influences social value partly through

mediation by transfer mechanisms. Technology transfer capability remains a dominant downstream driver and forms a critical linkage between innovation generation and application.

To visually present the contributions of different core driving variables across performance dimensions, see Figure 4a and Figure 4b. In the figures, the standardized impact strengths of the three core driving variables on efficiency, innovation, and outcome translation dimensions are presented as radar lines, clearly showing the differential effects across performance dimensions.

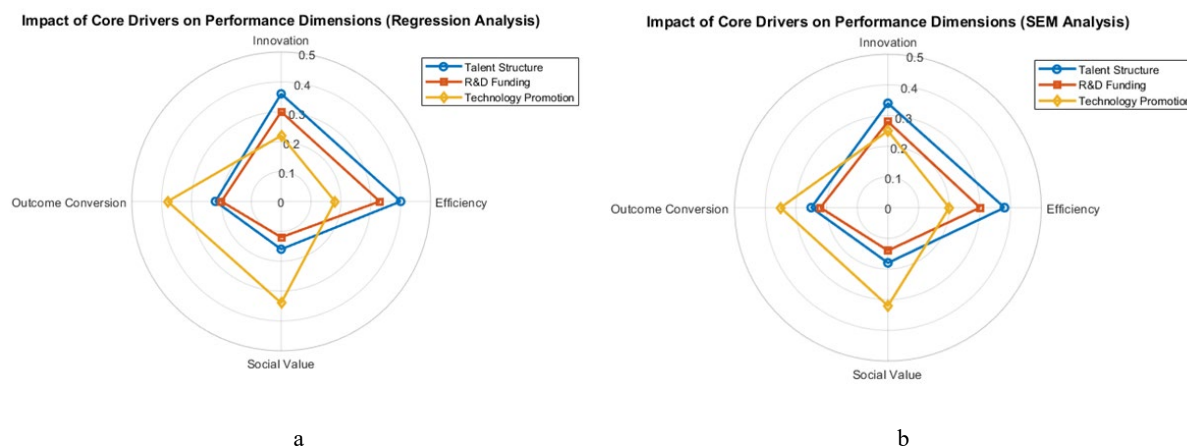


Figure 4. Impact of core drivers on performance dimensions: (a) regression analysis; (b) SEM analysis
Note: R&D = Research and Development; SEM = structural equation modeling.

Figure 4a illustrates the magnitude of the effects of core driving variables on each performance dimension. Human capital has the strongest impact on efficiency and innovation, while its contribution to outcome translation and social value is relatively low, indicating that the proportion of highly educated medical staff primarily drives improvements in research efficiency and innovation capacity. Research funding has a moderate impact on efficiency and innovation dimensions and also somewhat promotes outcome translation, but it has a limited effect on social value, reflecting the fundamental supporting role of financial investment in research activities. Technology transfer capability has the most pronounced effect on outcome translation and social value, while also positively influencing the innovation dimension, highlighting its importance as a pathway for translating research outcomes into clinical and societal applications. The regression analysis clearly reflects the differentiated roles of the three core driving variables across performance dimensions, providing empirical validation for the multi-objective optimization results.

Figure 4b shows the standardized path coefficients of core driving variables on performance dimensions obtained via SEM. Compared with regression analysis, SEM results further reveal the synergistic effects among driving variables. Human capital remains dominant in efficiency and innovation dimensions, but its influence on outcome translation is slightly enhanced, reflecting that human capital indirectly affects outcome application through collaborative mechanisms. Research funding maintains a positive effect on efficiency and innovation and also has indirect effects on outcome translation and social value, indicating that financial investment not only directly promotes research output but can also indirectly enhance value translation by optimizing resource allocation. Technology transfer capability has a significant impact on outcome translation and social value and also positively affects the innovation dimension through structural pathways, validating the importance of the “innovation-translation-application” closed-loop mechanism. The SEM results better capture the systemic and collaborative effects of core driving variables within the performance structure compared with regression analysis, offering more comprehensive guidance for policy-making and resource allocation.

Human capital and research funding mainly drive improvements in efficiency and innovation dimensions, whereas technology transfer capability plays a key role in outcome translation and social value dimensions. These findings provide empirical support for the multi-objective optimization results and quantitative guidance for policy design aimed at enhancing scientific and technological innovation performance in regional public hospitals.

3.3.3 Analysis of the effects of driving factors on different performance dimensions

To provide a system-level interpretation, Figure 5 summarizes how the identified drivers contribute differently across efficiency, innovation capability, and translation/social value dimensions. The output indicates that human capital primarily drives efficiency and innovation, while technology transfer capability dominates translation and social value performance. Research funding plays a bridging role across multiple dimensions, suggesting that funding allocation affects not only upstream productivity but also downstream translation readiness.

This cross-dimensional decomposition confirms that performance optimization cannot be achieved by

strengthening a single input dimension. Instead, it requires coordinated enhancement of upstream innovation capacity (human capital), midstream resource support (funding), and downstream diffusion capability (technology transfer).

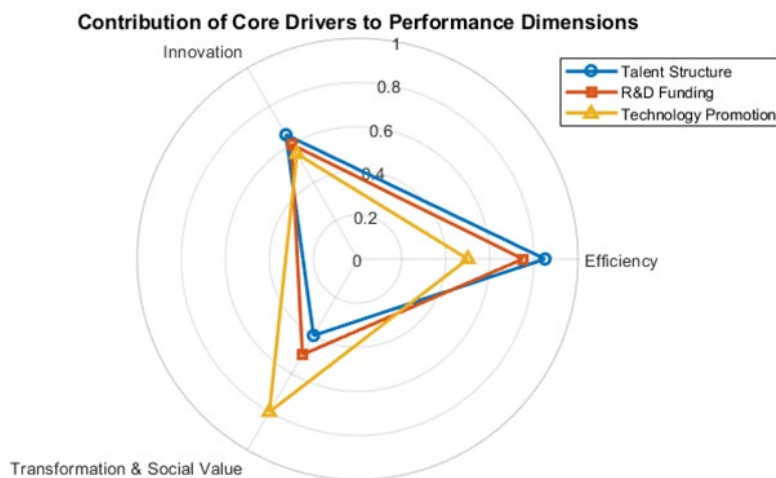


Figure 5. Contribution of core drivers to performance dimensions
Note: R&D = Research and Development.

As shown in the figure, human capital has the most significant effect on improving efficiency and innovation capability. An increase in the proportion of highly educated medical staff can substantially enhance research efficiency and innovation quality; however, its direct contribution to outcome translation and social value remains relatively limited. Research funding demonstrates a moderate-to-high contribution to both efficiency and outcome translation dimensions, and also provides certain support for innovation capability, reflecting its bridging role between research activities and outcome translation. In contrast, technology transfer capability contributes most strongly to the outcome translation and social value dimension, while also exerting some influence on innovation capability, whereas its direct effect on efficiency is comparatively weaker.

The three core driving variables show clear differences in their effects across performance dimensions: human capital primarily drives improvements in efficiency and innovation capability; research funding plays a bridging role across efficiency, innovation, and translation; and technology transfer capability mainly promotes outcome translation and the realization of social value. Through the visualization presented in Figure 5, the relative contributions of each driving variable across different dimensions can be clearly compared, providing theoretical justification and intuitive support for the subsequent design of implementation pathways under the multi-objective optimization framework.

3.4 Synergistic Mechanism Analysis

Beyond individual driver effects, the synergy interpretation module analyzes cross-dimensional coupling among efficiency, equity, and innovation value objectives. The synergy results are treated as structural outputs of the optimization process, explaining how indicator groups interact through complementary or substitutive mechanisms under constrained weight redistribution.

3.4.1 Synergistic relationships among efficiency, equity, and innovation value indicators

The coupling results suggest that efficiency and innovation value exhibit a positive synergy relationship, meaning that efficiency improvement provides upstream support for innovation output generation and translation readiness. Equity indicators function as structural constraints that reduce excessive concentration and improve distribution stability, thereby preventing efficiency gains from producing imbalanced downstream innovation value outcomes.

Figure 6 shows that the optimized system output produces a more balanced tri-dimensional configuration, indicating that the evaluation system is structurally reconfigured into a mutually supportive mechanism rather than an efficiency-dominant structure.

The figure shows that after optimization, the weights of efficiency, innovation value, and equity tend to become more balanced. The three dimensions form a mutually supportive structure, reflecting the synergistic logic under multi-objective balance. This provides empirical evidence for the structural optimization of the performance system and offers support for institutional design and policy formulation.

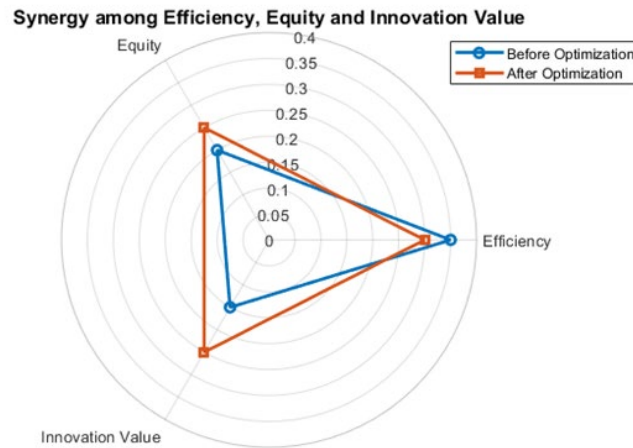


Figure 6. Synergy among efficiency, equity and innovation value

3.4.2 Complementarity and substitutability effects among indicators

The coupling analysis further indicates that indicator groups may exhibit complementarity or substitution under constrained optimization. When efficiency weights increase without synchronized enhancement of equity and innovation value indicators, the evaluation structure becomes output-biased and may intensify concentration effects, representing a substitution pattern. Conversely, when equity and innovation value weights increase, the system exhibits improved balance and sustainability, reflecting complementarity among objectives. Figure 7 visualizes these coupling patterns. Synchronous increases in multiple indicator groups indicate complementarity, whereas opposite movements reflect substitution-driven trade-offs.

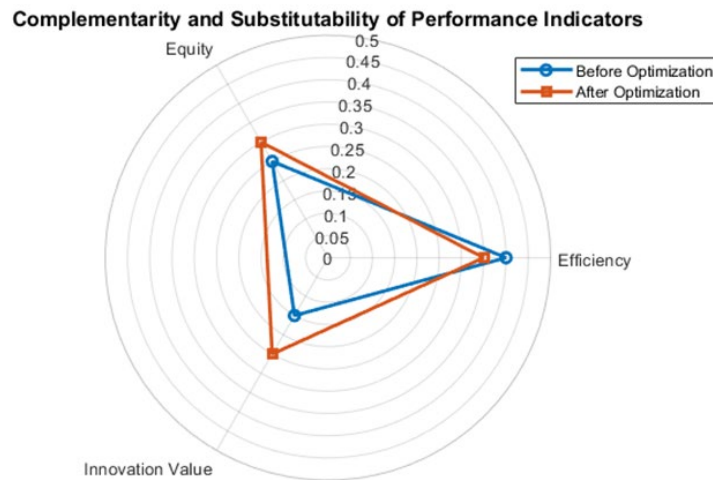


Figure 7. Complementarity and substitutability of performance indicators

3.4.3 Improvement of system stability and the balancing effect on resource allocation

From a system stability perspective, the optimized weight structure reduces excessive dispersion and decreases the probability of single-indicator dominance. This stabilizes evaluation outputs against metric fluctuations and enhances robustness of performance governance. Figure 8 shows that the optimized configuration produces a more evenly distributed weight structure, confirming that the model improves stability and balance simultaneously under multi-objective constraints.

Before optimization, the weights of certain dominant indicators were highly concentrated, making performance results sensitive to fluctuations in individual indicators. After optimization, the weight structure becomes more balanced, complementarity among indicators is strengthened, and the system responds more robustly to variations in resource allocation and innovation output. Figure 8 presents the changes in the weights of core indicators before and after optimization.

As shown in the figure, after optimization, the indicator weights tend to be more reasonably distributed, and the differences among efficiency indicators, innovation value indicators, and equity indicators are reduced, demonstrating a significant enhancement in resource allocation balance. This figure not only provides an intuitive

illustration of the improvement in system stability but also reveals the mechanism through which the optimization model achieves coordinated performance structure under multi-objective constraints, offering visual evidence for the design of subsequent implementation pathways.

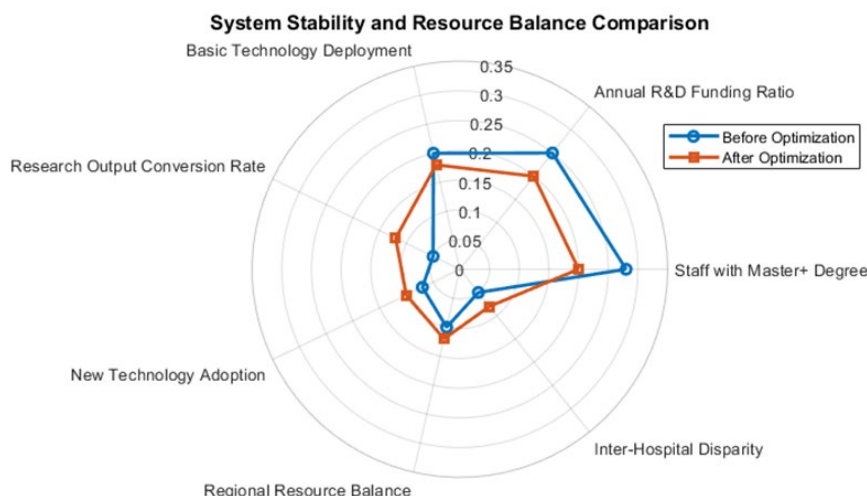


Figure 8. System stability and resource balance comparison
Note: R&D = Research and Development.

3.5 Subsample and Regional Heterogeneity Analysis

To test the generalizability of the decision-support model under heterogeneous contexts, subsample experiments are conducted by region (eastern, central, western) and hospital type (general, specialized, integrated). The results indicate that the optimization module consistently improves multi-objective balance across subsamples, but the redistribution magnitude differs due to baseline resource endowment and institutional specialization.

Eastern hospitals exhibit high baseline efficiency and translation readiness, resulting in limited marginal improvement in equity after optimization. In contrast, western hospitals show substantial improvements in both efficiency and equity, suggesting that the optimization module is more sensitive and beneficial under resource-scarce conditions. From a hospital type perspective, specialized hospitals gain more from translation-oriented weight reinforcement, while general hospitals show stronger efficiency gains but still require improvement in downstream translation capacity. Figure 9 visualizes the optimized weight outputs by region, indicating that the model generates differentiated structural configurations across heterogeneous contexts. These results suggest that the proposed framework can serve as an adaptive decision-support tool for region-specific governance tuning.

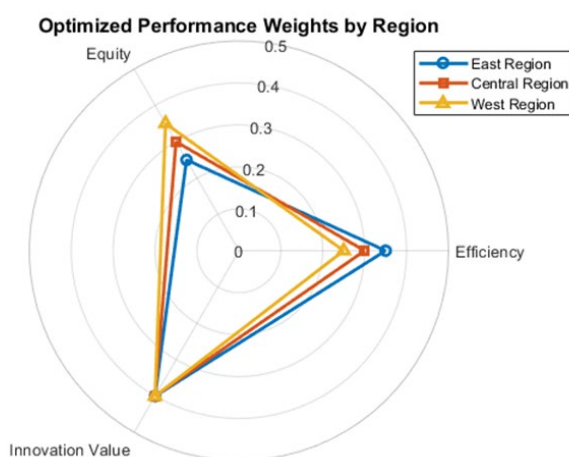


Figure 9. Optimized performance weights by region

In the figure, the polygons represent the contribution distributions of hospitals in the eastern, central, and western regions across the three dimensions of efficiency, equity, and innovation value. It can be clearly observed that all regions improve after optimization; however, differences in relative contributions across dimensions

remain, suggesting that performance optimization strategies should be designed in a differentiated manner according to regional characteristics. The subsample analysis confirms that the core driving factors—human capital, research funding, and technology transfer capability—are generally effective across different hospital types. However, their impact magnitudes and synergistic mechanisms vary by region and hospital type. These findings provide evidence for formulating region-specific and type-specific strategies to enhance scientific and technological innovation performance, and offer practical guidance for optimizing implementation pathways.

4. Implementation Pathway Design

Based on the optimization outputs and identified driver mechanisms, this section translates model results into operational decision-support strategies. Instead of providing macro-level policy recommendations, the proposed pathways are presented as implementable system modules that can be embedded into hospital innovation management processes, enabling dynamic weight adjustment, targeted resource scheduling, and translation-oriented governance tuning.

4.1 Dynamic Optimization Mechanism of Indicator Weights

In the process of optimizing scientific and technological innovation performance in regional public hospitals, the dynamic adjustment of indicator weights is a critical approach to achieving coordinated improvement across efficiency, equity, and innovation value. The multi-objective optimization results indicate that the innovation performance structure should shift from a single academic output orientation toward a synergistic orientation integrating efficiency, quality, and outcome translation. This requires the establishment of a flexible indicator weight adjustment mechanism in practice, rather than maintaining a long-term fixed academic-oriented weight structure.

Specifically, indicator weights for research efficiency, innovation quality, and outcome translation should be periodically revised based on the stage of regional innovation development and annual performance evaluation results, ensuring a dynamic balance among different objectives. Meanwhile, continuous monitoring of core driving indicators—such as human capital structure, research funding input, and technology transfer capability—should be strengthened to maintain their central roles within the innovation performance system. By establishing rolling adjustment and feedback mechanisms, indicator weights can be optimized in response to changes in innovation activities and resource conditions. This enables the performance evaluation system to evolve from a traditional static scoring tool into an instrument for structural optimization and management regulation, facilitating dynamic synergy among indicators and providing institutionalized support for innovation development in regional public hospitals.

4.2 Precision Allocation Mechanism of Innovation Resources

Based on the results of multi-objective optimization and driving factor analysis, the precise allocation of innovation resources is a key pathway for improving innovation performance in regional public hospitals. Empirical findings demonstrate that human capital and research funding significantly influence research efficiency, innovation quality, and outcome translation. Therefore, resource allocation should prioritize these core driving variables.

In practice, the structure of research funding should be optimized by increasing the proportion of financial support allocated to technology promotion and outcome translation processes, thereby strengthening the linkage between funding input and innovation value realization. In addition, mechanisms for the recruitment and cultivation of high-level talent should be improved to enhance the overall quality and sustainable innovation capacity of research teams. Furthermore, the quantitative results derived from the multi-objective optimization model can serve as a reference for resource allocation decisions, enabling the establishment of a performance feedback mechanism. This allows research funding and human resources to be reasonably directed toward high-contribution areas, improving efficiency while preventing excessive concentration of innovation resources. Such a precision allocation mechanism not only enhances resource utilization efficiency but also promotes synergistic improvement across efficiency, equity, and innovation value indicators, providing a solid institutional foundation for performance optimization.

4.3 Collaborative Promotion Mechanism for Innovation and Translation

The significant increase in the weight of outcome translation and application value in the multi-objective optimization results highlights the importance of extending innovation activities toward clinical practice and societal value. Accordingly, implementation efforts should strengthen the “innovation-translation-application” closed-loop mechanism. Specific measures include improving clinical translation channels for research outcomes,

establishing mechanisms for appropriate technology promotion and regional sharing, and enhancing the coverage of innovation outcomes within primary healthcare systems. Meanwhile, cross-institutional collaborative platforms should be developed to promote cooperative innovation among hospitals, universities, and research institutes, thereby improving resource integration efficiency.

Within the performance evaluation system, the weights of indicators related to outcome translation and application should be increased, guiding innovation activities to align more closely with clinical practice and regional health demands, and ensuring effective integration among research efficiency, innovation quality, and social value. Through this collaborative promotion mechanism, regional public hospitals can achieve dual improvements in scientific research innovation and social service capacity under institutionalized management, providing strong support for the sustainable development of innovation performance.

5. Conclusions

In addition to theoretical contributions, the proposed framework can be deployed as a data-driven decision-support tool within digital healthcare governance systems. The synergy interpretation module further provides interpretable structural evidence for balancing productivity, equity, and translation value. Such system-level integration enables regional health authorities and hospital administrators to implement adaptive governance tuning and improve innovation performance through quantifiable, model-driven decision outputs. Based on a multi-objective optimization framework, this study systematically investigates the driving factors, synergistic mechanisms, and structural optimization pathways of scientific and technological innovation performance in regional public hospitals. The findings indicate that performance improvement mainly relies on three core driving variables: human capital structure, research funding input, and technology transfer capability. Specifically, human capital—represented by the proportion of medical staff holding a master's degree or above—serves as a key factor in enhancing research efficiency and innovation quality. Research funding input not only supports the implementation of research activities but also directly influences outcome translation and technology promotion. Technology transfer capability plays a decisive role in transforming innovation achievements into clinical applications and social services. These results suggest that the scientific allocation of key innovation resources and the optimization of human capital structure constitute the core pathway for improving innovation performance.

Regarding synergistic mechanisms, the indicators of efficiency, equity, and innovation value exhibit a dynamic coordination relationship. The multi-objective optimization results demonstrate that under resource constraints, effective balancing across performance dimensions can be achieved through weight reallocation and indicator structure optimization, thereby avoiding performance bias caused by the dominance of a single indicator. Efficiency indicators enhance the input-output efficiency of research investment, equity indicators mitigate imbalances in innovation resource distribution across regions and hospitals, and the optimization of innovation value indicators significantly strengthens outcome translation and contributions to social services. Through this synergistic mechanism, the innovation performance system not only achieves structural optimization but also improves overall system stability and long-term sustainability.

Based on the analysis of driving factors and synergistic mechanisms, this study proposes practical implementation pathways. First, a dynamic optimization mechanism for indicator weights should be established, enabling periodic adjustments of efficiency, equity, and innovation value weights according to the stage of regional innovation development and annual performance evaluation results, thereby ensuring dynamic balance within the multi-objective structure. Second, a precision allocation mechanism for innovation resources should be implemented by rationally allocating research funding and optimizing strategies for the recruitment and cultivation of high-level talent, improving the effectiveness of resource allocation and enhancing the efficiency of innovation outcome translation. Third, an innovation-translation-application closed-loop mechanism should be constructed by improving channels for technology promotion and outcome application, promoting the extensive adoption of innovation achievements in primary healthcare systems and regional social services, and ultimately achieving the integrated alignment of performance improvement and social value objectives.

Although this study provides important findings, several limitations remain. First, the sample only covers the regional public hospitals included in the original dataset, which may limit the generalizability of the conclusions. Second, the multi-objective optimization model assumes linear contributions among indicators and does not fully account for potential nonlinear coupling effects. Third, certain weight parameters still involve subjectivity, which may influence the interpretation of results. In addition, the study period is relatively limited, and the dynamic characteristics of performance changes have not been tracked over a long-term horizon.

Future research can be extended in several directions. First, dynamic multi-objective optimization methods can be introduced to enable adaptive weight adjustment in response to policy changes and hospital development stages. Second, DEA, machine learning, and other methods can be integrated to assign indicator weights more objectively, thereby improving model interpretability and robustness. Third, the sample size can be expanded by incorporating cross-regional and multi-period data to validate model applicability under different environmental and temporal conditions. Fourth, nonlinear coupling relationships and potential interaction effects among indicators can be

explored to provide more refined theoretical and methodological support for optimizing innovation performance evaluation systems.

This study not only identifies the core driving factors of scientific and technological innovation performance in regional public hospitals and clarifies their underlying mechanisms, but also proposes operational policy and management pathways. The findings provide systematic guidance for improving hospital innovation capability, optimizing resource allocation, and promoting the translation of innovation outcomes, offering important theoretical support and practical implications for enhancing governance models of innovation performance in regional public hospitals.

Author Contributions

Conceptualization, Y.Y.W. and X.L.; methodology, L.H.Z.; software, Z.Q.L.; validation, Z.Q.L., Y.Y.W., and L.H.Z.; formal analysis, Y.Y.W.; investigation, X.L.; resources, X.L.; data curation, X.L.; writing—original draft preparation, Y.Y.W.; writing—review and editing, L.H.Z.; visualization, Z.Q.L.; supervision, X.L.; project administration, Y.Y.W.; funding acquisition, Y.Y.W. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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